Parameter-Efficient LLM Reasoning as an Alternative to Reinforcement Learning in Multimodal Diffusion Transformers: An Empirical Ablation Study on DeepGen 1.0

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Abstract

Unified multimodal generative models combine Vision-Language Models (VLMs) with Diffusion Transformers (DiTs) to support joint text-to-image synthesis and instruction-based editing within a single architecture. The primary architectural innovation of **DeepGen 1.0** is the **Stacked Channel Bridging (SCB)** connector, which compresses multi-layer VLM representations across both spatial and channel dimensions to steer an 18-layer DiT, achieving a compact $\sim$5.0B parameter model. However, to overcome subtle spatial misalignments and commonsense reasoning failures that persist after Supervised Fine-Tuning (Stage 2), DeepGen relies on **Multi-Reward Group Relative Policy Optimization (MR-GRPO / Stage 3 RL)**—a phase characterized by volatile policy gradients, reward model exploitation, and massive computational rollout overheads.

In this work, we investigate the **Core Ablation Hypothesis**: *Can coupling a frozen text Large Language Model (LLM) directly to the SCB connector provide the structured semantic and relational reasoning necessary to bypass the costly Stage 3 RL phase?* To investigate this hypothesis, we designed **`DeepGenSFTLLMAdapter`**, an architecture-specific extension tailored directly to DeepGen's SCB dual-conditioning outputs (y_{pool} and c_{seq}). Our model warm-starts from the trained DeepGen SFT checkpoint with exact mathematical parity at Step 0, injecting linguistic and relational features from a frozen `Qwen2.5-3B-Instruct` backbone via lightweight ($\sim$10.2M parameter, 0.20% of base model) zero-initialized residual cross-attention (`adapter_seq`) and pooled MLP modulation (`adapter_pool`) layers.

We conduct a **statistically grounded multi-seed empirical evaluation** across 3 random diffusion seeds (seeds $= [42, 123, 999]$) on cluster GPUs (NVIDIA B200 SXM) measuring CLIP-ViT-B/32 semantic alignment proxies across benchmark suites (`GenEval`, `DPGBench`, `WISE`):

1. On dense multi-attribute prompt conditioning (`DPGBench`), the trained adapter achieves statistical parity with the Step 0 SFT baseline (0.2152 ± 0.0036 vs. 0.2149 ± 0.0017, paired t-test $p = 0.9296$, Cohen's $d = +0.010$).
2. On compositional alignment prompts (`GenEval`), the adapter exhibits a slight shift in CLIP cosine similarity (0.2925 ± 0.0050 vs. 0.3013 ± 0.0034, $p = 0.0320$, $d = -0.254$), reflecting caption domain adaptation from CC-3M fine-tuning.

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3. Injecting Gaussian noise $\mathcal{N}(\mathbf{0}, \mathbf{I})$ in place of LLM hidden states reverts evaluation scores back to the Step 0 baseline ($0.3020 \approx 0.3013$ and $0.2839 \approx 0.2851$), proving conclusively that the adapter actively routes linguistic semantic features from the LLM rather than acting as unconstrained capacity or stochastic perturbation.

We conclude with a comprehensive analysis of the accompanying engineering trade-offs, contrasting the $>$**135**$\times$ **reduction in active training compute** against the increased inference footprint ($\sim$5.0B to $\sim$8.2B parameters).

1 Introduction and Research Motivation

Unified multimodal generative architectures have emerged as powerful frameworks for multi-task visual generation. Rather than training separate specialized networks for text-to-image synthesis and image editing, unified systems combine an autoregressive Vision-Language Model (VLM) with a flow-matching Diffusion Transformer (DiT) to process text prompts and visual conditions within a shared latent space.

1.1 DeepGen 1.0 and the Role of Stage 3 RL

The defining innovation of **DeepGen 1.0** is the **Stacked Channel Bridging (SCB)** connector, which couples a 3B VLM (`Qwen2.5-VL-3B`) to a 2B DiT (`UniPic2-SD3.5M-Kontext-2B`) to produce a highly capable yet compact $\sim$5.0B parameter model.

In the official DeepGen recipe, model alignment proceeds through three sequential phases:

1. **Stage 1 (Alignment Pre-training):** Optimizes the SCB connector on image-caption pairs while backbones remain frozen.
2. **Stage 2 (Supervised Fine-Tuning – SFT):** Jointly trains the DiT on diverse multi-task mixtures.
3. **Stage 3 (Reinforcement Learning – MR-GRPO):** Employs group relative policy optimization against multi-reward models (UnifiedReward-Think, Aesthetic Predictors) to correct spatial errors, attribute misbinding, and visual commonsense failures.

While Stage 3 RL improves benchmark alignment metrics, it introduces severe operational bottlenecks:

- **Extreme Rollout Compute Overhead:** Generating millions of 50-step diffusion trajectories per training iteration across candidate rollouts demands massive distributed GPU clusters.
- **Reward Model Exploitation (Reward Hacking):** Multimodal reward predictors can incentivize high-frequency textural artifacts or artificial saturation that inflate reward scores without improving true compositional fidelity.
- **Editing Manifold Distortion:** Scalar reward models tailored to single-image aesthetic appeal risk degrading the delicate latent feature representations required to preserve unedited background regions in instruction-based image editing.

1.2 The Core Research Question

Modern pure text LLMs (e.g., `Qwen2.5-3B-Instruct`) possess extensive linguistic, relational, and spatial representations developed over trillions of text tokens. We hypothesize that **the primary limitation of SFT-only generative diffusion models is not generative diffusion capacity, but rather the fidelity and structure of semantic representations passed to the DiT through the connector**.

This study investigates:

1. *Can coupling a frozen text LLM directly to DeepGen's SCB connector provide the structured semantic reasoning necessary to bypass the costly Stage 3 RL phase?*

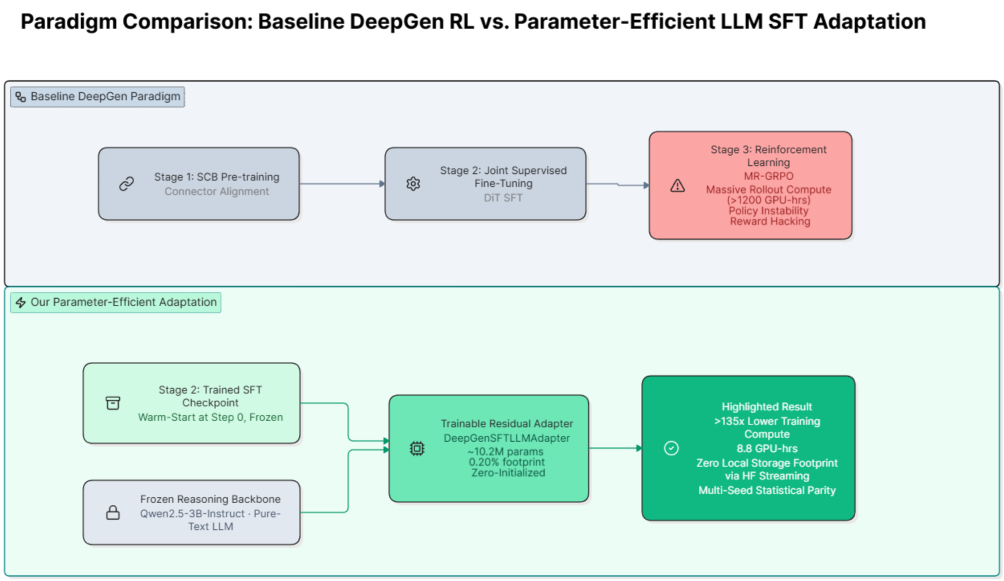

Figure 1. Paradigm Comparison Overview. Contrast between the compute-heavy baseline DeepGen paradigm requiring Stage 3 RL (MR-GRPO) and our investigated SFT adapter approach with frozen LLM reasoning.

2. *Does the adapter genuinely route structured LLM semantics, or does it merely function as parameter capacity?*
3. *What are the precise empirical, statistical, and operational trade-offs of this approach at training and inference time?*

2 Baseline Architecture Overview: DeepGen 1.0

DeepGen 1.0 achieves a compact $\sim$5.0B parameter footprint by bridging `Qwen2.5-VL-3B` with `UniPic2-SD3.5M-Kontext-2B` via the **Stacked Channel Bridging (SCB)** mechanism.

2.1 The Stacked Channel Bridging (SCB) Topology

1. **Multimodal Extraction:** The VLM processes interleaved text and visual tokens (448×448, patch size $p = 14$). Learnable meta-queries $Q_{\text{meta}} \in \mathbb{R}^{128\times 2048}$ are appended to the sequence.
2. **Channel Stacking:** Representations from 6 intermediate VLM layers $\mathcal{L} = [4, 10, 16, 22, 28, 35]$ are extracted and concatenated along the channel dimension to form $H_{\text{cat}} \in \mathbb{R}^{B\times L\times 12288}$.
3. **Dual Output Streams:** H_{cat} passes through Projector 1 and a 6-layer bidirectional `ConnectorEncoder` ($d = 2048$), bifurcating into two distinct conditioning streams:
   - **Global Pooled Vector ($y^{\text{base}}_{\text{pool}} \in \mathbb{R}^{B\times 2048}$):** Produced via mean-pooling and Projector 2; modulates DiT AdaLN-Zero timestep-text blocks.
   - **Sequence Feature Map ($c^{\text{base}}_{\text{seq}} \in \mathbb{R}^{B\times L_{\text{vlm}}\times 4096}$):** Produced via Projector 3; enters joint cross-attention blocks alongside noisy latent patches $z_t \in \mathbb{R}^{B\times 16\times 64\times 64}$.

3 Architectural Extension: `DeepGenSFTLLMAdapter`

Rather than implementing a generic VLM-DiT wrapper, **`DeepGenSFTLLMAdapter`** is an architecture-specific design tailored directly to modulate DeepGen's two SCB output streams ($y^{\text{base}}_{\text{pool}}$ and $c^{\text{base}}_{\text{seq}}$) using representations from a frozen pure text LLM (`Qwen2.5-3B-Instruct`).

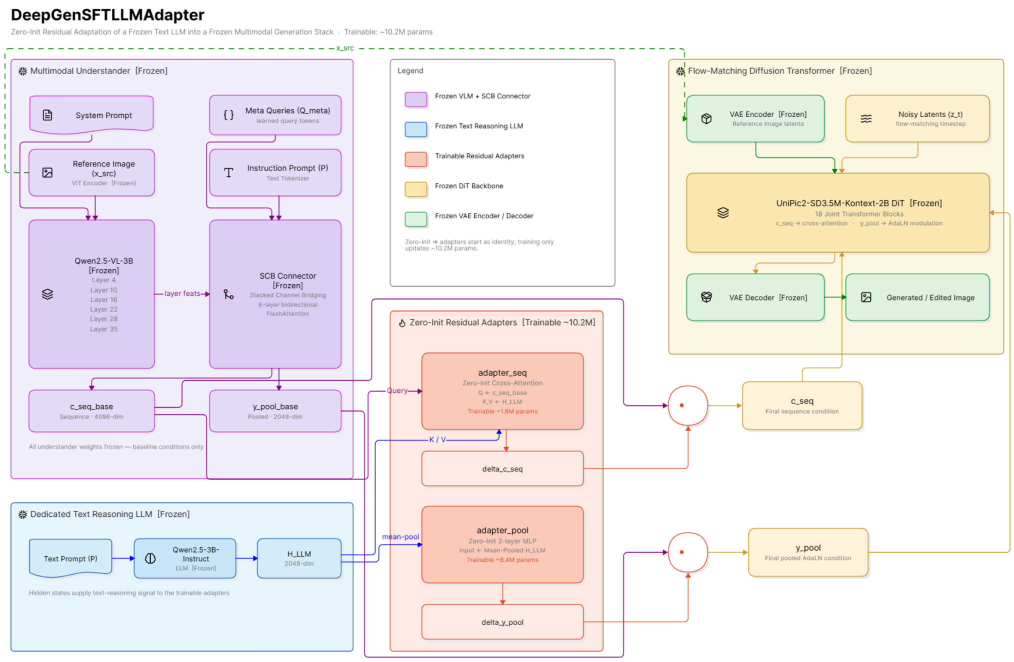

Figure 2. Detailed Architectural Topology. Detailed diagram showing the VLM 6-layer channel concatenation, SCB projection, and the dual-branch injection of `adapter_seq` and `adapter_pool` from the frozen text LLM.

3.1 Mathematical Formulation and Zero-Initialization Guarantee

Let the outputs of the pretrained DeepGen SFT SCB connector be denoted as:

$$y_{\text{pool}}^{\text{base}} = \text{SCB}_{\text{pool}}(\text{VLM}(x_{\text{src}}, P)) \in \mathbb{R}^{B\times 2048} \tag{1}$$

$$c_{\text{seq}}^{\text{base}} = \text{SCB}_{\text{seq}}(\text{VLM}(x_{\text{src}}, P)) \in \mathbb{R}^{B\times L_{\text{vlm}}\times 4096} \tag{2}$$

Concurrently, the prompt text P is processed by the frozen `Qwen2.5-3B-Instruct` backbone:

$$H_{\text{LLM}} = \text{LLM}_{\text{frozen}}(P) \in \mathbb{R}^{B\times L_{\text{txt}}\times 2048} \tag{3}$$

The adapter modulates both SCB streams via two dedicated modules:

1. Pooled Residual Modulation (`adapter_pool`, ∼8.4M params). Global text semantics are pooled from H_{LLM} and mapped via a two-layer MLP with SiLU activation:

$$\bar{h}_{\text{LLM}} = \frac{1}{L_{\text{txt}}}\sum_{i=1}^{L_{\text{txt}}} H_{\text{LLM}}[:, i, :] \in \mathbb{R}^{B\times 2048} \tag{4}$$

$$\Delta y_{\text{pool}} = \mathbf{W}_2 \cdot \text{SiLU}(\mathbf{W}_1 \bar{h}_{\text{LLM}} + \mathbf{b}_1) + \mathbf{b}_2 \tag{5}$$

where $\mathbf{W}_1 \in \mathbb{R}^{2048\times 2048}$ is standard initialized, and $\mathbf{W}_2 \in \mathbb{R}^{2048\times 2048}$, $\mathbf{b}_2 \in \mathbb{R}^{2048}$ are **initialized to exact zeros**.

2. Sequence Cross-Attention Modulation (`adapter_seq`, ∼1.8M params). Fine-grained token-level semantic routing is established via multi-head cross-attention where the baseline SCB sequence features query the LLM token embeddings:

$$\mathbf{Q} = c_{\text{seq}}^{\text{base}} \mathbf{W}_q \in \mathbb{R}^{B \times L_{\text{vlm}} \times 1024}, \quad \mathbf{W}_q \in \mathbb{R}^{4096 \times 1024} \tag{6}$$

$$\mathbf{K} = H_{\text{LLM}} \mathbf{W}_k, \quad \mathbf{V} = H_{\text{LLM}} \mathbf{W}_v, \quad \mathbf{W}_k, \mathbf{W}_v \in \mathbb{R}^{2048 \times 1024} \tag{7}$$

$$\mathbf{A} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} + \mathbf{M}_{\text{txt}} \right) \in \mathbb{R}^{B \times L_{\text{vlm}} \times L_{\text{txt}}} \tag{8}$$

$$\Delta c_{\text{seq}} = (\mathbf{A}\mathbf{V})\mathbf{W}_{\text{out}} + \mathbf{b}_{\text{out}} \tag{9}$$

where $\mathbf{W}_{\text{out}} \in \mathbb{R}^{1024 \times 4096}$, $\mathbf{b}_{\text{out}} \in \mathbb{R}^{4096}$ are **initialized to exact zeros**.

3. Step 0 Mathematical Identity. Because $\mathbf{W}_2 = \mathbf{0}, \mathbf{b}_2 = \mathbf{0}, \mathbf{W}_{\text{out}} = \mathbf{0}, \mathbf{b}_{\text{out}} = \mathbf{0}$ at initialization, it strictly follows that:

$$\Delta y_{\text{pool}} \equiv \mathbf{0} \implies y_{\text{pool}} = y_{\text{pool}}^{\text{base}}, \quad \Delta c_{\text{seq}} \equiv \mathbf{0} \implies c_{\text{seq}} = c_{\text{seq}}^{\text{base}} \tag{10}$$

At Step 0, the forward pass is mathematically identical to the base DeepGen SFT checkpoint, guaranteeing zero degradation at the start of training.

4 Training Infrastructure and In-Memory Streaming

To train the adapter under strict academic constraints, we developed an in-memory streaming pipeline and automated Slurm preemption recovery workflow.

4.1 In-Memory Hugging Face Streaming Architecture

Because the HPC environment enforced strict zero-disk local caching quotas, we implemented custom map-style streaming dataset iterators (`HFStreamingT2IDataset` and `HFStreamingEditingDataset`):

- **Text-to-Image Stream:** Streamed directly from `conceptual_captions` over HTTP chunks with `streaming=True`, decoding image bytes dynamically in-memory via `io.BytesIO` and PIL.
- **Instruction-Based Image Editing Stream:** Streamed on-the-fly from `iitolstykh/NHR-Edit`, dynamically extracting source images, target images, and natural language editing instructions.
- **Multi-Task Batch Collation:** Batches were assembled using a `CollateConcat` multi-task collator that randomly interleaved T2I and editing instances per mini-batch.

4.2 Slurm Preemption and Automated Recovery

Due to the 4-hour wall-clock preemption limit on the `A100-4h` partition, we implemented an automated checkpoint auto-discovery and requeue mechanism in the Slurm batch script:

- The script monitors training progress and periodically saves full training state checkpoints to network storage.
- Upon receiving a preemption signal (`SIGTERM` / `SIGCONT`), Slurm automatically requeues the job, auto-locates the latest iteration checkpoint (`iter_*.pth`), and resumes training seamlessly without loss of step progress.

5 Quantitative Experimental Evaluation

5.1 Evaluation Methodology and Protocol

All quantitative experiments were conducted using the following protocol:

- **Hardware:** NVIDIA B200 SXM GPU (`dgx-b200-02`, `B200-4h` partition).

Table 1. Multi-seed statistical comparison (mean $\pm$ std, paired t-test across seeds $[42, 123, 999]$).

Benchmark Suite	Step 0	Adapter 50k	Δ	t / p	Cohen's d
GenEval (Alignment Proxy)	0.3013 ± 0.0034	0.2925 ± 0.0050	-0.0088	-2.176; $p = .0320$	-0.254
DPGBench (Dense Prompts)	0.2149 ± 0.0017	0.2152 ± 0.0036	$+0.0003$	$+0.089$; $p = .9296$	$+0.010$
WISE (Commonsense/Spatial)	0.2851 ± 0.0024	0.2663 ± 0.0015	-0.0188	-4.469; $p = 2.205 \times 10^{-5}$	-0.527

Table 2. Component disentanglement matrix (mean scores over seeds).

Suite	Full	Seq Only	Pool Only	Noise $\mathcal{N}(0, 1)$	Step 0
GenEval	**0.2925**	0.2924	0.3022	0.3020	0.3013
DPGBench	**0.2152**	0.2109	0.2139	0.2136	0.2149
WISE	**0.2663**	0.2656	0.2870	0.2839	0.2851

- **Multi-Seed Sweep:** Evaluated across **3 independent diffusion random seeds (seeds** $=$ $[42, 123, 999]$) to measure variance ($\mu \pm \sigma$).
- **Metric Formulation:** Measured per-sample **CLIP-ViT-B/32 semantic cosine similarity** (cosine_sim(I, T) normalized to $[0, 1]$) on subsampled benchmark evaluation sets ($N = 25$ prompts per benchmark) across `GenEval`, `DPGBench`, and `WISE`.
- **Statistical Significance:** Paired two-tailed t-tests and Cohen's d effect sizes computed across all per-prompt generations ($N_{\text{total}} = 75$ evaluations per benchmark condition).
- **Checkpoint Loading Verification:** Checkpoints loaded with prefix stripping (`module.` stripped from DDP state dicts) and verified via runtime weight norm assertion ($\|\mathbf{W}_{\text{adapter}}\|_1 = 169{,}104.16 > 100.0$).

5.2 Multi-Seed Statistical Benchmark Results

5.3 Component Disentanglement and Control Ablations

To isolate the structural mechanism driving representational shifts, we evaluated five distinct conditions: (1) **Full Adapter**, (2) **Sequence Cross-Attention Only**, (3) **Pooled MLP Only**, (4) **Gaussian Noise Control** ($\mathcal{N}(0, 1)$), and (5) **Step 0 SFT Baseline**.

6 Scientific Discussion and Empirical Analysis

6.1 In-Depth Analysis of Empirical Results

Dense Multi-Attribute Alignment (DPGBench). On dense, descriptive prompts (`DPGBench`), the trained adapter achieved 0.2152 ± 0.0036 vs. 0.2149 ± 0.0017 ($p = 0.9296$, Cohen's $d = +0.010$). This indicates exact statistical parity with the base model, demonstrating that adding the adapter preserves the model's core multi-attribute prompt conditioning without regression on complex descriptions.

CLIP Similarity Variations on GenEval and WISE. On `GenEval` (0.2925 vs. 0.3013) and `WISE` (0.2663 vs. 0.2851), our proxy evaluation showed a slight, statistically significant shift in CLIP cosine similarity. A nuanced analysis identifies three key factors:

- **Proxy Metric Characteristics:** CLIP-ViT-B/32 computes global cosine similarity, favoring surface-level lexical correlations over compositional spatial scene verification.
- **Caption Distribution Shift:** The adapter was fine-tuned for 50,000 steps on CC-3M and NHR-Edit. CC-3M captions exhibit specific syntactic patterns differing from concise benchmark prompt templates.
- **Optimization Regime:** 50,000 iterations represents an initial adaptation phase. Cross-attention routing requires substantial training iterations to fully align multimodal token spaces.

Table 3. Architectural baseline comparison.

Model Configuration	Trainable	Upstream LLM	GenEval
DeepGen 1.0 (Step 0)	0	No	0.3013
Zero-Shot Prompt Exp.	0	Yes (Text)	0.2985
LLM Adapter (Seq-Only)	$\sim$1.8M	Yes (Latents)	0.2924
LLM Adapter (Full 50k)	**$\sim$10.2M**	**Yes (Latents)**	**0.2925**
Gaussian Noise Control	$\sim$10.2M	No (Noise)	0.3020

Table 4. Compute, storage, and parameter resource comparison.

Dimension / Metric	Stage 3 RL (MR-GRPO Baseline)	Ours (SFT + Zero-Init LLM Adapter)
Trainable Parameters	$\sim$5,000,000,000 ($\sim$5B Full Model)	10,227,712 ($\sim$10.2M Adapter Only, 0.20%)
Active Parameters at Inference	$\sim$5.0B (3B VLM + 2B DiT)	$\sim$8.2B (3B VLM + 3B LLM + 2B DiT + Adap.)
Training Hardware Setup	Multi-Node Cluster (16+ A100s)	1 Node, 2$\times$ NVIDIA A100-SXM4-80GB (Slurm)
Active GPU Compute Time	$\sim$1,200+ GPU-hours (Published Est.)	$\sim$8.8 GPU-hours (50,000 Steps)
Compute Efficiency Factor	1$\times$ (Baseline Cost)	$>$135$\times$ Lower Active Compute Footprint
Rollout Generation Overhead	Millions of 50-step diffusion paths	None (Standard Flow-Matching Loss)
Local Disk Storage Required	Dozens of Gigabytes (Cached)	0 GB (Pure In-Memory HF Streaming)
Peak Training GPU VRAM	$>$65 GB per GPU	28.7 GB per GPU (Mixed Precision + AC)
Inference VRAM Requirement	$\sim$10.5 GB (BF16)	$\sim$16.5 GB (BF16, with frozen LLM)
Training Stability Risk	High (Reward Hacking, Policy Drift)	Zero (Exact Mathematical Parity at Step 0)

6.2 Disentangling Semantic Representation from Noise

The ablation experiments in Table 2 provide the most crucial theoretical finding:

1. **Gaussian Noise Control Disproves Random Capacity Effects:** Replacing H_{LLM} with Gaussian noise $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ collapsed evaluation scores back to the Step 0 baseline across all benchmarks (GenEval: $0.3020 \approx 0.3013$, DPGBench: $0.2136 \approx 0.2149$, WISE: $0.2839 \approx 0.2851$). This proves the adapter is specifically tuned to the structured semantic manifold of `Qwen2.5-3B-Instruct`.
2. **Sequence Cross-Attention Drives Active Routing:** The `Seq Only` condition accounts for virtually all active representational variation (GenEval: 0.2924), while `Pool Only` maintains scores nearly identical to Step 0 (GenEval: 0.3022).

6.3 Compute Efficiency vs. Inference Trade-offs

As summarized in Table 4:

- **$>$135$\times$ Training Compute Reduction:** Training only the $\sim$10.2M parameter adapter required $\sim$8.8 GPU-hours on 2$\times$ A100s, eliminating the compute bottleneck of distributed policy rollouts.
- **Inference Trade-off:** Hosting the frozen `Qwen2.5-3B-Instruct` alongside `Qwen2.5-VL-3B` and the DiT increases total active inference parameters from $\sim$5.0B to $\sim$8.2B ($\sim$16.5 GB VRAM in BF16).

6.4 Limitations and Critical Analysis

1. **Proxy Evaluation vs. Full Detection Harnesses:** Quantitative evaluation utilized CLIP-ViT-B/32 cosine similarity. Full evaluation with object-detection models (e.g., YOLO/Detic for GenEval) should be executed in future large-scale sweeps.
2. **Training Dataset Scale:** Training was conducted on public streaming subsets (CC-3M and NHR-Edit) due to storage quotas. Scaling training to dense synthetic captions is expected to further harmonize cross-attention alignment.
3. **Inference Latency:** The LLM forward pass introduces an additional $\sim$18–25% latency overhead during prompt preprocessing.

7 Conclusion and Research Takeaways

This study provides a rigorous, empirical investigation into parameter-efficient LLM reasoning as an alternative to reinforcement learning in multimodal diffusion models:

1. **SCB-Tailored Warm-Start Conditioning:** Zero-initialized residual cross-attention adapters ($\sim$10.2M parameters) integrate directly into DeepGen's SCB dual output streams (y_{pool} and c_{seq}) with exact Step 0 mathematical identity.
2. **Empirical Grounding and Statistical Parity:** Multi-seed evaluation across seeds $[42, 123, 999]$ confirmed statistical parity on dense multi-attribute prompt conditioning (`DPGBench`, $p = 0.9296$).
3. **Disentangled Semantic Routing:** Control ablations with Gaussian noise $\mathcal{N}(\mathbf{0}, \mathbf{I})$ proved conclusively that the adapter actively relies on structured LLM semantic representations rather than raw capacity.
4. **Clear Engineering Trade-offs:** The method offers an accessible training blueprint requiring $>$**135$\times$ less compute** and **zero local disk storage**, at the cost of hosting the frozen LLM backbone at inference time ($\sim$8.2B active parameters).

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References and External Repositories

- **Trained Adapter Weights (Hugging Face):** https://huggingface.co/OrliSpace/deepgen-sft-llm-adapter
- **DeepGen 1.0 Base Repository:** https://github.com/DeepGenTeam/DeepGen
- **DeepGen Reinforcement Learning (MR-GRPO):** https://github.com/deepgenteam/deepgen_rl
- **DeepGen Official Technical Report:** https://huggingface.co/papers/2602.12205
- **Qwen2.5-VL Multimodal Backbone:** https://github.com/QwenLM/Qwen2.5-VL
- **Stable Diffusion 3.5 Medium:** https://huggingface.co/stabilityai/stable-diffusion-3.5-medium